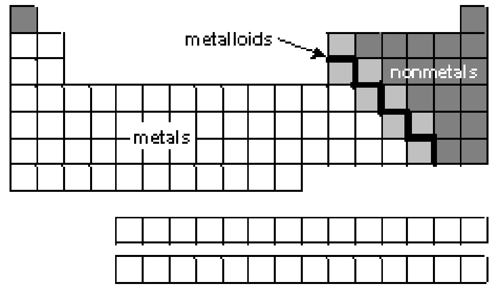
NAME ______________________________Period _____ Test Date_______________

Unit 5 –The Periodic Table: Essential Knowledge and Guided Notes

Read Textbook: CK Chemistry Chapter 6: The Periodic Table
https://flexbooks.ck12.org/cbook/ck-12-chemistry-flexbook-2.0/
Watch Helpful Videos and Practice: Posted on CANVAS
Quick Electron Configuration Practice: **http://tinyurl.com/geekconfig**
Fun Videos: Four New Elements:https://tinyurl.com/4newelements
Will the Periodic Table be Complete? https://tinyurl.com/ptcomplete
Crash Course Chemistry: https://tinyurl.com/crashcoursept



Highlight Essential Knowledge:

1. **Mendeleev** organized his Periodic Table by atomic mass. **Mosley** correctly organized the Periodic Table by atomic number.
2. The significance of the Periodic Law:
   a. **The Periodic Law** states that when elements are arranged in order of increasing atomic numbers, their physical and chemical properties show a periodic pattern.
   b. **Periodicity** is regularly repeating patterns or trends in the chemical and physical properties of the elements arranged in the periodic table.

3. The Periodic Table consists of three main types of elements. The properties of **metals**, **nonmetals,** and **metalloids** differ.

4. Metals are good conductors of heat and electricity, are ductile (can be drawn into wires), and malleable (can be pounded into sheets). Most metals are solid at room temperature. Metals form cations by losing electrons to have a full outer energy level.
   This is called **oxidation**: $M \rightarrow M^{+1}$ + electron (e^{1-})
   Example: $Na \rightarrow Na^{1+} + e^{1-}$
5. The most reactive metals are Group 1 metals, followed by Group 2 metals and transition metals. Metals with larger atomic masses as you move down a group are increasingly more reactive. Cesium (and Francium) are the most reactive metals on the periodic table.

6. Nonmetals are poor or non-conductors of heat and electricity and they are brittle if they are solid. Many nonmetals are gases. One is a liquid (Bromine). Non-metals form anions by gaining electrons to have a full outer energy level.
   This is called **reduction:** NM + electron (e^{1-}) $\rightarrow NM^{1-}$
   Example: $F + e^{1-} \rightarrow F^{1-}$
7. The most reactive non-metals are Group 17 nonmetals, followed by the Group 16 non-metals Non-metals with smaller atomic masses as you move up a group are increasingly more reactive. Fluorine is the most reactive non-metal on the periodic table. Group 18 gases are "inert" meaning very non-reactive due to full valence electron energy levels.

8. Metalloids also called semi-metals include Boron, Silicon, Arsenic, Germanium, Antimony, Tellurium, and Polonium and have some properties of both metals and non-metals. For example, Silicon is a semiconductor.

9. Periods and groups are named by numbering columns and rows. Horizontal rows called **periods** have predictable properties based on an increasing number of electrons in the outer energy levels. Vertical columns, called **groups or families**, have similar properties because of their similar valence electron configurations. Four groups have common "family names". The Transition metals are located in the middle of the periodic table and have incomplete d sublevels.

10. Ions differ from neutral elements as they have lost or gained electrons to obtain a noble gas configuration. Elements in the same group produce the same ions.

Column (Family)	**Valence Electrons**	**Electron Configuration of Valence Electrons**	**Common Charge or Oxidation State**
1 (alkali metals)	1	ns^1	+1
2 (alkaline earth metals)	2	ns^2	+2
3-12 (transition metals)	Varies (Usually 1,2,3 or 4)	Varies	Usually +1,+2 or +3
13	3	ns^2np^1	+3
14	4	ns^2np^2	Non-metals and metalloids usually do not form ions. Sn and Pb +2 or +4
15	5	ns^2np^3	-3 for non-metals in the group
16	6	ns^2np^4	-2 for non-metals in the group
17 (halogens)	7	ns^2np^5	-1
18 (noble gases)	8 (except He has 2)	ns^2np^6 except He (ns^2)	Do not form ions

11. There are trends in the periodic table that relate to ionization energy, electronegativity, shielding effect, and atomic sizes as measured by the atomic radius.

    a. **The trend for atomic size (radius)**: Atomic radius is the measure of the distance between the radii of two identical atoms of an element. Atomic radius decreases from left to right and increases from top to bottom within given groups.
    b. **The trend for the shielding effect**: Electron shielding refers to the blocking of valence shell electron attraction by the nucleus, due to the presence of inner-shell electrons. The shielding effect is constant within a given period and increases within given groups from top to bottom.
    c. **The trend for ionization energy**: Ionization energy is the energy required to remove the most loosely held electron from a neutral atom. Ionization energies generally increase from left to right and decrease from top to bottom of a given group.
    d. **The trend for electronegativity**: Electronegativity is the measure of the attraction of an atom for electrons in a bond. Electronegativity increases from left to right within a period and decreases from top to bottom within a group. The trend in ionization energy is very similar to Electronegativity, except for the Noble Gases that have no Electronegativity but very high Ionization Energies.

12. The **ionic radius** is the size of an ion.
    a. The ionic radius of a metal cation is smaller than the metal atom because in losing an electron they lose an occupied energy level.
    b. The ionic radius of a non-metal anion is larger than the non-metal atom because by gaining an electron they increase the electron repulsion in the occupied energy level causing it to expand.

U5 Lesson #1 - Periodic Table Basics

Part A: Organization

1. Who proposed the first periodic table? ________________________________
2. Who proposed the first periodic table that was arranged by atomic number? _____________________________________
3. The Periodic Table shows all known __________________________ in the universe. It organizes elements by _____________________________________.
4. A great deal of information about an element can be gathered from its ______________ on the periodic table. For example, you can predict with reasonably good accuracy the physical and chemical properties of the element. You can also predict what other elements a particular element will ___________________________
5. Fill in the table:

Metals	Non-Metals	Metalloids
• _________ conductors of heat and electricity • All are ___________ at room temperature • High luster and sheen • Ductile and malleability	• _______ conductors of heat and electricity • Most are ____________ at room temperature • Physical properties vary greatly • Usually brittle	• Properties are similar to both ________________ and ________________

6. **The Periodic Law:**
   a. When elements are arranged in order of increasing atomic number, there is a ____________________________ of their physical and chemical properties.
   b. The elements within the same ______________ have similar properties. This is because they have the same number of ________________________________.
   c. The elements within the same _____________ repeat predictably as you move from one period to the next.
   d. The columns are called ______________ and represent the ________________ of the last electron of each element in that column
   e. The rows are called _________________ and represent the ________________ of the element's valence electrons that reside in that row.
7. **Groups on the Periodic Table:**

Column #	Common Name	# of Valence Electrons	Metal/NonMetal/Metalloid or Combination?
1			
2			
3-12			
13			
14			
15			
16			
17			
18			

Part B: Electron Configurations of Ions

8. How do atoms achieve greater stability? (What does every atom want?)
   a. To have a _______________________ energy level.

   b. To look like a ______________________.

   c. Atoms form _____________ to achieve this.

   d. They want to achieve full s and p orbitals. This is called the _______________.

   e. The most common __________________ or most common ________________

      __________ is where the atom has a full outer energy level.

   f. Metals ______ electrons becoming _________. This is called ______________.

   g. Non-metals _____electrons becoming ________. This is called _____________.

9. Behavior of Metals: From your electron configuration of these metals, write the most common ion charge by removing the valence electrons until you have the configuration of a noble gas

Atom	Na	Mg	Al
Electron Configuration	[Ne]$3s^1$	[Ne]$3s^2$	[Ne]$3s^2 3p^1$
# Valence Electrons			
# Electrons lost or gained to get a full energy level			
Ion (charge and value)			
What noble gas does this ion behave like?			
Oxidation or Reduction?			

Oxidation is __________of an electron (e^{1-}) to form a cation.
Atom → Ion with charge + # of valence electrons lost

Na → __________ + _____e^{1-}

Mg→ __________ + _____e^{1-}

Al → __________ + _____e^{1-}

10. Behavior of Nonmetals: From your electron configuration of these nonmetals, write the most common ion charge by adding more valence electrons until you have the configuration of a noble gas

Atom	N	O	F
Electron Configuration	$[He]2s^2 2p^3$	$[He]2s^2 2p^4$	$[He]2s^2 2p^5$
# Valence Electrons			
# Electrons lost or gained to get a full energy level			
Ion (charge and value)			
What noble gas does this ion behave like?			
Oxidation or Reduction?			

Reduction is __________of an electron (e^{1-}) to form an anion.
Atom + # of valence electrons gained → Ion with charge

N + _____e^{1-} → __________

O + _____e^{1-} → __________

F + _____e^{1-} → __________

11. Common oxidation state (charge) for the following groups of elements:

Group 1 = ______

Group 2 = ______

Group 13 (metals only) = ______

Group 15 (nonmetals only) = ______

Group 16 (nonmetals only) = ______

Group 17 = ______

Group 18 = ______

12. Ways to remember oxidation vs reduction:

- **OIL RIG** = _________________________ is loss of electrons, ______________________ is gain of electrons.
- **LEO lion says GER** = Lose electrons is ________________, gain electrons is __________________

Unit 5 Lesson #2 - Periodic Trends

Part A: Reactivity and Family Characteristics

The ______________law states that when elements are arranged in order of increasing atomic numbers, their physical and chemical properties show a periodic pattern.

____________________is regularly repeating patterns or trends in the chemical and physical properties of the elements arranged in the periodic table.

Columns on the Periodic Table are called Families or Groups because they have the same number of ____________________________________. Families act in similar ways.

Metal "Families"

_________ Metals (Group 1)	_______________ ____________ Metals (Group 2)	____________________ Metals (Groups 3-12)
• Shiny and soft • Very reactive • 1 valence electron • They are not found in nature as a free element, but generally as salts. • When exposed to air, they tarnish due to oxidation. • They react when coming into contact with water. Some of them will even explode when they come into contact with water.	• Silvery, shiny, and relatively soft • Fairly reactive • 2 valence electrons • They all occur in nature but are only found in compounds and minerals, not in their elemental forms.	• Form many compounds with different oxidation states • Form compounds with different colors • Good conductors of electricity • High melting and boiling points • Relatively high densities

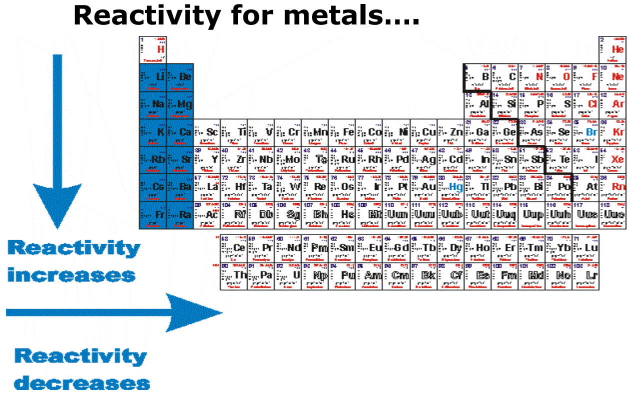
Metals ______________________ in reactivity as you go down these groups and _____________________ as you move from left to right across the row.

Non-Metal "Families"

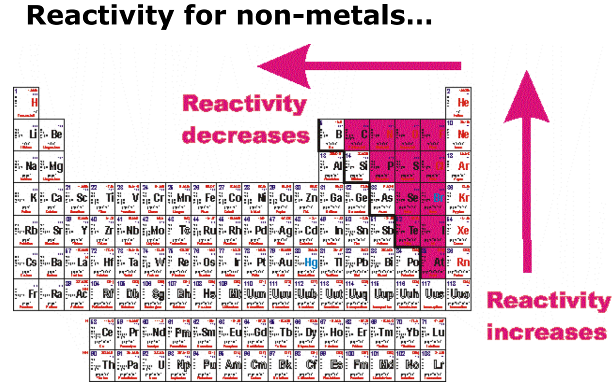
___________________ (Group 17)	____________ __________ (Group 18)
• Form acids when combined with hydrogen (HCl - hydrochloric acid) • Toxic • 7 valence electrons • Highly reactive and electronegative	• A full outer shell of electrons (Helium has 2 valence electrons and the rest have 8) • Very inert and stable • Usually don't tend to react with other elements to form compounds • Colorless and odorless gases

Non-Metals ___________________ in reactivity as you go up these groups and ______________________ as you move from right to left.

Reactivity for metals....



Reactivity for non-metals...



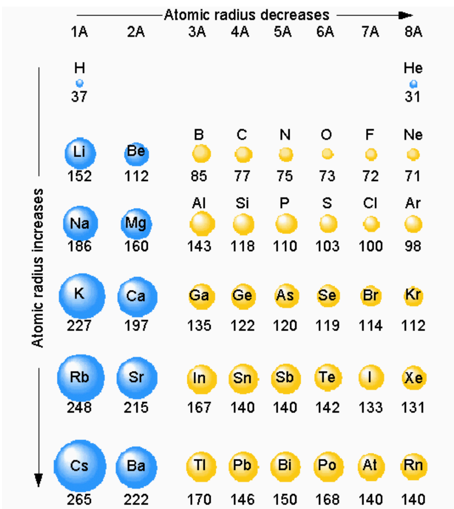
Part B: 4 Important Periodic Trends

Trend: Atomic Radius (a measure of ___________)

Definition: Half the distance between the nuclei of two identical atoms bonded together.

The atomic radius of atoms generally ______________________ from left to right across a period. THE INCREASED NUCLEAR CHARGE ATTRACTS THE ELECTRONS MORE STRONGLY.

The atomic radius of atoms generally ______________________ from top to bottom within a group. VALENCE ELECTRONS ARE FOUND ON HIGHER ENERGY LEVELS FURTHER FROM THE NUCLEUS.



Practice Questions:

1. Largest Radius? Li, C, or F?____

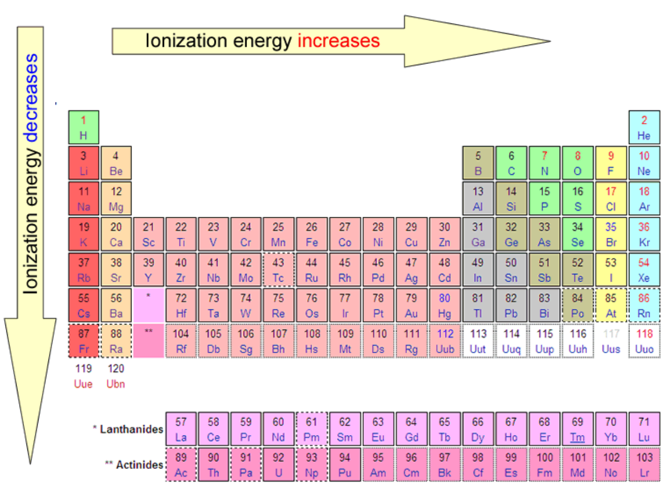
2. Smallest Radius? Li, Na, K, or Cs? ______

Trend: Ionization Energy (a measure of the strength of attraction between the nucleus and the _______________ electrons)

Definition: the energy required to remove an electron from an atom (to make a +1 ion)

The ionization energy of atoms generally ____________________ from left to right across a period. THE INCREASED NUCLEAR CHARGE ATTRACTS THE ELECTRONS MORE STRONGLY.

The ionization energy of atoms generally ___________________ from top to bottom within a group. VALENCE ELECTRONS ARE FOUND ON HIGHER ENERGY LEVELS FARTHER FROM THE NUCLEUS.



Practice Questions:

1. Largest IE? Li, C, or F?____

2. Smallest IE? Li, Na, K, or Cs? _____

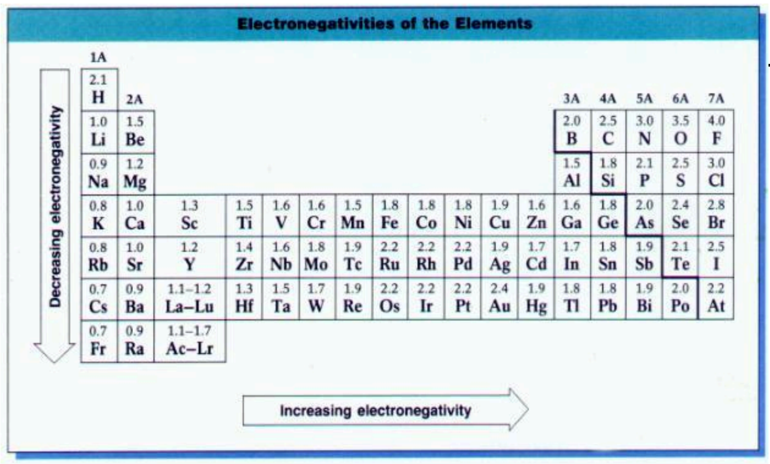
Trend: Electronegativity (a measure of the strength of attraction between the nucleus and the outside "_______________" electrons.)

Definition: the measure of the ability of an atom to attract electrons in a chemical bond.

The electronegativity of atoms generally _____________________ from left to right across a period.
THE INCREASED NUCLEAR CHARGE ATTRACTS THE ELECTRONS MORE STRONGLY

The electronegativity of atoms generally ________________ from top to bottom within a group.
"BONDING" ELECTRONS ARE FOUND ON HIGHER ENERGY LEVELS FARTHER FROM THE NUCLEUS

The ___________ ______________ generally do not have electronegativity as they have no room to attract electrons (full valence shells).



Electronegativities of the Elements

1A	2A											3A	4A	5A	6A	7A
2.1 H																
1.0 Li	1.5 Be											2.0 B	2.5 C	3.0 N	3.5 O	4.0 F
0.9 Na	1.2 Mg											1.5 Al	1.8 Si	2.1 P	2.5 S	3.0 Cl
0.8 K	1.0 Ca	1.3 Sc	1.5 Ti	1.6 V	1.6 Cr	1.5 Mn	1.8 Fe	1.8 Co	1.8 Ni	1.9 Cu	1.6 Zn	1.6 Ga	1.8 Ge	2.0 As	2.4 Se	2.8 Br
0.8 Rb	1.0 Sr	1.2 Y	1.4 Zr	1.6 Nb	1.8 Mo	1.9 Tc	2.2 Ru	2.2 Rh	2.2 Pd	1.9 Ag	1.7 Cd	1.7 In	1.8 Sn	1.9 Sb	2.1 Te	2.5 I
0.7 Cs	0.9 Ba	1.1–1.2 La–Lu	1.3 Hf	1.5 Ta	1.7 W	1.9 Re	2.2 Os	2.2 Ir	2.2 Pt	2.4 Au	1.9 Hg	1.8 Tl	1.8 Pb	1.9 Bi	2.0 Po	2.2 At
0.7 Fr	0.9 Ra	1.1–1.7 Ac–Lr														

Practice Questions:

1. Largest EN? Li, C, or F? ______

2. Smallest EN? Li, Na, K, or Cs? ___

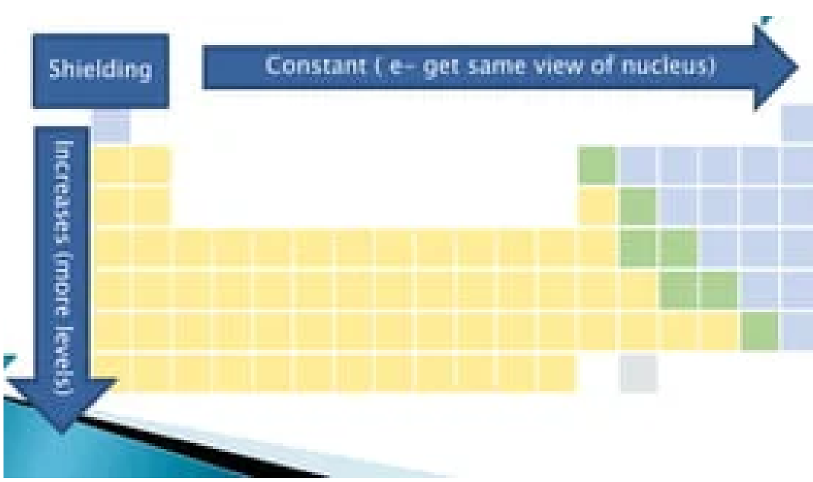
Trend: Shielding Effect

Definition: Shielding occurs when the inner electrons in an atom shield the outer electrons from the full charge of the nucleus.

The shielding effect of atoms generally _________________________ from left to right across a period.
ATOMS IN A PERIOD HAVE THE SAME NUMBER OF INNER (CORE) ELECTRONS

The shielding effect of atoms generally ______________________ from top to bottom within a group.
THE VALENCE ELECTRONS ARE IN HIGHER ENERGY LEVELS (MORE INNER/CORE ELECTRONS)

Keep in mind that this phenomenon is only important as you move down the periodic table!

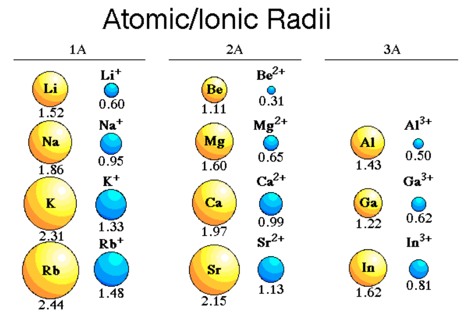


Practice Questions:

1. Which has greater shielding Na or Cl? __________

2. Which has greater shielding Cl or I? ___________

Comparing Atomic and Ionic Radii



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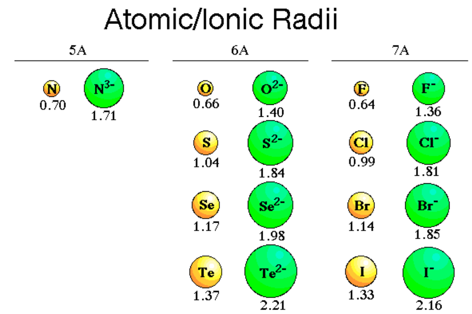
- Positively charged ions formed when an atom of a metal **loses one or more electrons**
- **Smaller** than the parent atom due to **loss of an energy level**

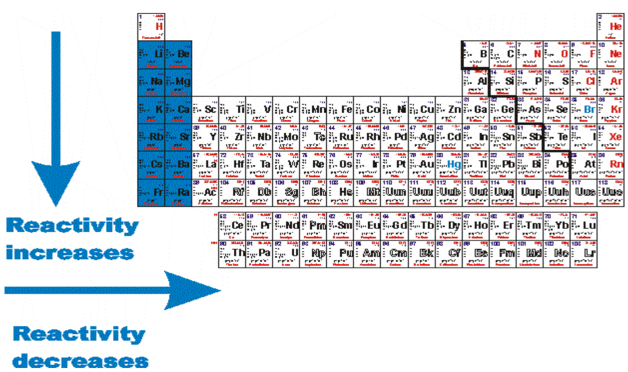


____

- Negatively charged ions formed when nonmetallic atoms **gain one or more electrons**
- **Larger** than the parent atom due to **repulsion of more electrons** in the same energy level

TRENDS TO KNOW!

Reactivity for metals....



Reactivity for non-metals...